

Solution: $p: n=abc$ $q: a \mid \sqrt[3]{n}$ $r: b \mid \sqrt[3]{n}$ $s: c \mid \sqrt[3]{n}$

We need to prove $p \rightarrow (q \vee r \vee s)$ is True. We shall give a proof by contraposition. $\therefore$ We shall show $\neg(q \vee r \vee s) \rightarrow \neg p$ is True.

Let, $\neg(q \vee r \vee s)$ be True. By De Morgan's Law, $(\neg q \wedge \neg r \wedge \neg s)$ is True.

By Defn. of AND, $\neg q$ is True, $\neg r$ is True and $\neg s$ is True.

$\neg p: n \neq abc$ $\neg q: a \nmid \sqrt[3]{n}$ $\neg r: b \nmid \sqrt[3]{n}$ $\neg s: c \nmid \sqrt[3]{n}$

$\therefore a \nmid \sqrt[3]{n}, b \nmid \sqrt[3]{n}, c \nmid \sqrt[3]{n} \therefore abc \nmid \sqrt[3]{n} \cdot \sqrt[3]{n} \cdot \sqrt[3]{n} = n$

$\therefore n \neq abc$. $\neg p$ is True. By Defn. of conditional, $\neg(q \vee r \vee s) \rightarrow \neg p$ is True. $\therefore p \rightarrow (q \vee r \vee s)$ is True $\square$

36. Prove that $\sqrt[3]{2}$ is irrational.

Solution: Suppose, for the sake of contradiction, $\sqrt[3]{2}$ is rational. By Defn. 2, $\exists$ integers a and b in lowest terms such that $\sqrt[3]{2} = \frac{a}{b}$. $\therefore \frac{a^3}{b^3} = 2 \Rightarrow a^3 = 2b^3$.

By Defn 1, a^3 is even.

Claim: For any arbitrary integer n , if n^3 is even, then n is even.

Proof: $p: n^3$ is even. $q: n$ is even

$\neg p: n^3$ is not even, i.e. n^3 is odd (By Defn. 1)

$\neg q: n$ is not even, i.e. n is odd (By Defn. 1)

We shall prove that $(p \rightarrow q)$ is True by contraposition.

$\therefore$ We shall prove $(\neg q \rightarrow \neg p)$ is True.

Let, $\neg q$ be True. $\therefore n$ is odd. By Defn. 1, $\exists$ integer K s.t. $n = (2K+1)$

$n^3 = (2K+1)^3 = 8K^3 + 12K^2 + 6K + 1 = 2(4K^3 + 6K^2 + 3K) + 1$.

It's easy to see that $(4K^3 + 6K^2 + 3K)$ is an integer. By Defn. 1, n^3 is odd.

$\therefore \neg p$ is True. By defn. of conditional, $(\neg q \rightarrow \neg p)$ is True. $\therefore (p \rightarrow q)$ is True. $\square$